

Pin Name	Mode / Function	Hardware Connection	Description
PA0	ADC1_IN0	CCD Pin 5 (via 1k/4.7k divider)	Green Signal Input
PA1	ADC1_IN1	CCD Pin 17 (via 1k/4.7k divider)	Red Signal Input
PA4	ADC1_IN4	CCD Pin 18 (via 1k/4.7k divider)	Blue Signal Input
PA8	TIM1_CH1	MIC4427 Input A	$\Phi 1$ (Master Clock)
PA9	TIM1_CH2	MIC4427 Input B	$\Phi 2$ (Inverted Clock)
PA10	TIM1_CH3	CCD Pin 3 (via Buffer)	Φ RS (Reset Pulse)
PB12	GPIO_Output	CCD Pins 8, 9, 14 (via Buffer)	Φ ROG (Trigger)
PB0	GPIO_Output	Green LED	Status: Scanning Green now
PB1	GPIO_Output	Blue LED	Status: Scanning Blue now
PB2	GPIO_Output	Red LED	Status: Scanning Red now
PB13	GPIO_Output	ORANGE LED (New!)	WARNING: Aflatoxin Found
PC13	GPIO_Output	Onboard LED	System Alive (Solid ON)